

LECTURE 11: CONFIDENCE SETS. BAYESIANS IN LARGE SAMPLES.

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REFERENCE This is not really covered in HMC, apart from some basics in Chapter 4.2. Chapter 9 in CB is a much better reference. H22, Chapter 14 and 16.13.

In this lecture, we will study confidence sets. We observe data X , with distribution $F(\cdot \mid \theta)$, $\theta \in \Theta$. The parameter space Θ may be finite- or infinite-dimensional. We are interested in summarizing what we know about the parameter of interest $g(\theta)$ by constructing a data-dependent set $C(X)$ that contains $g(\theta)$ with high probability.

Definition 1. A *confidence set* for a parameter $g(\theta)$ is any data-dependent set $C(X) \subseteq g(\Theta)$. The *coverage probability* of a confidence set $C(X)$ for $g(\theta)$ is the probability of containing $g(\theta)$, $P_\theta(g(\theta) \in C(X))$. The minimal coverage probability, $\inf_{\theta \in \Theta} P_\theta(g(\theta) \in C(X))$, is called the *confidence coefficient*. A confidence set $C(X)$ for a parameter $g(\theta)$ is said to have *confidence level* $1 - \alpha$ if the confidence coefficient is at least $1 - \alpha$, $\inf_{\theta \in \Theta} P_\theta(g(\theta) \in C(X)) \geq 1 - \alpha$. If $C(X)$ takes the form of an interval, we call it a *confidence interval (CI)* with level $1 - \alpha$.

Often, the term *nominal level* is used to describe the confidence level that the confidence set hopes to achieve, and the term *true level* or *actual level* is used to mean the confidence coefficient. For particular values of θ , the coverage probability of the set $C(X)$ may be higher than the confidence level, just as the null rejection rate of a test may be lower than its significance level for particular null values of θ . Note that the randomness in the coverage comes from the data X , in contrast to a Bayesian credible set, where the randomness comes from θ .

In standard problems, if $g(\theta)$ is a scalar, the set $C(X)$ will often take the form of an interval, $[L(X), U(X)]$. One possibility is to set $L(X) = -\infty$ and $U(X) = \infty$. Such an interval will contain $g(\theta)$ with probability 1, a large confidence level indeed. Of course, the problem with this interval is that it is not informative about $g(\theta)$. So, we want the set $C(X)$ to simultaneously achieve two things: have a small volume (or be short, if it's an interval), and also contain $g(\theta)$ with high probability.

In principle, one should set up the problem as a decision problem to spell out the tradeoff between volume and coverage, because it's actually not so obvious how one

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should resolve it. We will not do so here; instead, we follow the custom in the literature to seek sets with small volume, subject to a confidence level constraint.

Next, we consider the most common method for finding confidence sets. We then consider its large-sample analog. Finally, we will compare confidence sets to Bayesian credible intervals.

1. TEST INVERSION

For each $g(\theta) \in g(\Theta)$, consider the problem of testing $H_0: g(\theta) = g_0$ against $H_1: g(\theta) \neq g_0$. Suppose that for each such hypothesis we have a non-randomized test $\phi_{g_0}(X)$ with significance level α . Then the confidence set collecting all the nulls that were not rejected,

$$C(X) = \{g_0 \in g(\Theta): \phi_{g_0}(X) = 0\}$$

has confidence level $1 - \alpha$. Indeed, since the test $\phi_{g(\theta)}$ has level α by construction, for any $\theta \in \Theta$,

$$P_\theta(g(\theta) \in C(X)) = 1 - E_\theta[\phi_{g(\theta)}(X)] \geq 1 - \alpha,$$

which implies $\inf_{\theta \in \Theta} P_\theta(g(\theta) \in C(X)) \geq 1 - \alpha$. This procedure is called *test inversion*. Test inversion can produce confidence sets that are non-convex—in general, there is no guarantee we end up with an interval.

Conversely, if we have a way of constructing a confidence set with confidence level $1 - \alpha$, we can use it for testing $H_0: g(\theta) = g_0$ against a two-sided alternative by accepting the null whenever $g_0 \in C(X): \phi_{g_0}(X) = \mathbb{1}\{g_0 \notin C(X)\}$. This test will have level α .

Example 1. Let $X = (X_1, \dots, X_n)$ be a random sample from distribution $\mathcal{N}(\theta, \sigma^2)$, σ^2 known. Let us use test inversion to construct a confidence set for θ of level $1 - \alpha$. Consider the problem of testing the null hypothesis, $H_0: \theta = \theta_0$ against the alternative, $H_1: \theta \neq \theta_0$. The uniformly most powerful (UMP) unbiased test rejects if $|\bar{X}_n - \theta_0| \geq z_{1-\alpha/2}\sigma/\sqrt{n}$. Thus, $\theta \in C(X)$ iff

$$-z_{1-\alpha/2}\sigma/\sqrt{n} \leq \bar{X}_n - \theta \leq z_{1-\alpha/2}\sigma/\sqrt{n}.$$

Rearranging the expression yields the interval

$$[\bar{X}_n - z_{1-\alpha/2}\sigma/\sqrt{n}, \bar{X}_n + z_{1-\alpha/2}\sigma/\sqrt{n}]. \quad (1)$$

So in this case, we actually get an interval. Consider inverting the one-sided z-test of the null $H_0: \theta = \theta_0$ against the alternative $H_1: \theta > \theta_0$, which accepts if $\bar{X}_n - \theta_0 \leq z_{1-\alpha}\sigma/\sqrt{n}$. In this case, $\theta \in C(X)$ iff

$$\bar{X}_n - z_{1-\alpha}\sigma/\sqrt{n} \leq \theta,$$

which yields the one-sided CI, $[\bar{X}_n - z_{1-\alpha}\sigma/\sqrt{n}, \infty)$. $\square$

Example 2. Let $X = (X_1, \dots, X_n)$ be a random sample from the distribution $\mathcal{N}(\mu, \sigma^2)$, with $\theta = (\mu, \sigma^2)$. Let us first use test inversion to construct a confidence set for σ^2 . Consider the problem of testing the null hypothesis, $H_0: \sigma^2 = \sigma_0^2$ against the alternative, $H_1: \sigma^2 \neq \sigma_0^2$. Under the null hypothesis, $(n-1)S_n^2/\sigma_0^2 \sim \chi^2(n-1)$, so this is a pivot. Therefore, if we accept the null whenever

$$F_{\chi^2(n-1)}^{-1}(\alpha/2) \leq (n-1)S_n^2/\sigma_0^2 \leq F_{\chi^2(n-1)}^{-1}(1-\alpha/2),$$

where $F_{\chi^2(n-1)}^{-1}(\alpha)$ is the α quantile of the $\chi^2(n-1)$ distribution, the test will have level α . Rearranging, we accept all nulls $H_0: \sigma^2 = \sigma_0^2$ that satisfy

$$(n-1)S_n^2/F_{\chi^2(n-1)}^{-1}(1-\alpha/2) \leq \sigma_0^2 \leq (n-1)S_n^2/F_{\chi^2(n-1)}^{-1}(\alpha/2).$$

Again, the confidence set is an interval, which has coverage $1-\alpha$ independently of the value of μ . Notice that the quantiles get switched: the upper quantile determines the lower endpoint of the CI.

We have also seen that for testing the null $H_0: \mu = \mu_0$, the t -statistic, $T_n = (\bar{X}_n - \mu_0)/\sqrt{S_n^2/n} \sim t(n-1)$, is a pivot. Inverting the t -statistic gives us the usual CI, $\bar{X}_n \pm t_{1-\alpha/2}(n-1)S_n/\sqrt{n}$. $\boxtimes$

2. PRATT'S THEOREM (OPTIONAL)

Intuitively, inverting powerful tests should yield short confidence intervals (confidence sets with small volume). Inverting tests that are UMP against a particular alternative θ_1 should yield confidence intervals that are as short as possible when θ_1 is the true parameter. The next result formalizes this intuition.

Theorem 1 (Pratt 1961). Suppose X has density $f(x | \theta)$, $\theta \in \Theta \subseteq \mathbb{R}$. Let $C(X)$ be a confidence set for θ based on inverting tests $\phi_\theta(X)$, and let $\lambda(C(X)) = \int_\Theta (1 - \phi_\theta(X)) d\theta$ denote its volume. Then, for any θ_1 ,

$$E_{\theta_1}[\lambda(C(X))] = \int_\Theta (1 - \pi_{\phi_\theta}(\theta_1)) d\theta = \int_\Theta P_{\theta_1}(\theta \in C(X)) d\theta,$$

where $\pi_{\phi_\theta}(\theta_1) = E_{\theta_1}[\phi_\theta(X)]$ denotes the power of the test ϕ_θ against θ_1 . Furthermore, if the tests $\phi_\theta(X)$ are UMP against θ_1 , and the expected length is finite, then $C(X)$ achieves the shortest expected length at θ_1 among all confidence sets with level $1-\alpha$.

Proof. The first result follows from changing the order of integration:

$$E_{\theta_1}[\lambda(C(X))] = E_{\theta_1} \left[\int_\Theta (1 - \phi_\theta(X)) d\theta \right] = \int_\Theta \int_{\mathcal{X}} (1 - \phi_\theta(x)) f(x | \theta_1) dx d\theta = \int_\Theta (1 - \pi_{\phi_\theta}(\theta_1)) d\theta.$$

If $C(X)$ is based on inverting tests ϕ_θ that are UMP against θ_1 , then for each $\theta \neq \theta_1$, such tests minimize $1 - \pi_{\phi_\theta}(\theta_1)$, and therefore also minimize the integral. $\square$

The second part, although often cited to justify the practice of constructing confidence sets by inverting tests, is not that useful since it only tells us about how to minimize the length of CI at a particular point, and often, this comes at the expense of terrible expected length at other points (generically, we won't simultaneously be able to minimize length at all θ).

Remark 1. The current theorem is useful, however, as a benchmark for the best possible performance of a CI in more complicated problems: if we construct a CI that is “close”, in some sense, to the Pratt bound uniformly over θ , then we know one cannot improve much over such CI.

Example 1 (continued). Since the CI in Example 1 is based on inverting UMP unbiased tests, the CI in eq. (1) is the CI with the shortest expected length among all intervals based on inverting unbiased tests. However, stated in this way, it's not clear why one should restrict attention to unbiased tests.

What if we don't restrict attention to unbiased tests? Then we can use Theorem 1 as a benchmark for the best possible performance of any CI at a given θ_1 . Specifically, by Theorem 1, the confidence set with the shortest expected length when $\theta = \theta_1$ is given by inverting UMP tests of the null $H_0: \theta = \theta_0$ against the alternative $H_1: \theta = \theta_1$. By the Neyman-Pearson lemma, these are one-sided z-tests that reject if $\bar{X}_n - \theta_0 > \zeta$ if $\theta_0 < \theta_1$, and reject if $\bar{X}_n - \theta_0 < -\zeta$ if $\theta_0 > \theta_1$, where $\zeta = z_{1-\alpha}\sigma/\sqrt{n}$. This means that the CI consists of all values of θ that either satisfy $(\bar{X}_n - \zeta \leq \theta \text{ and } \theta \leq \theta_1)$, or else satisfy $(\bar{X}_n + \zeta \geq \theta \text{ and } \theta \geq \theta_1)$. We may write this as

$$\{\theta: \bar{X}_n - \zeta \leq \theta \leq \theta_1 \quad \text{or} \quad \theta_1 \leq \theta \leq \bar{X}_n + \zeta\}$$

Based on the value of $\bar{X}_n$, it may be the case that either both sets of inequalities hold at once, or else only one holds. Breaking down the possible cases, we obtain the CI

$$\begin{cases} [\bar{X}_n - \zeta, \theta_1] & \text{if } \bar{X}_n + \zeta \leq \theta_1, \\ [\bar{X}_n - \zeta, \bar{X}_n + \zeta] & \text{if } \bar{X}_n - \zeta \leq \theta_1 \leq \bar{X}_n + \zeta, \\ [\theta_1, \bar{X}_n + \zeta] & \text{if } \bar{X}_n - \zeta \geq \theta_1. \end{cases} \quad (2)$$

It then follows by some algebra (fill it in!) that the expected length of the CI when $\theta = \theta_1$ is

$$\frac{2\sigma}{\sqrt{n}} [z_{1-\alpha}(1 - 2\alpha) + E[(Z + z_{1-\alpha}) \mathbb{1}\{Z > z_{1-\alpha}\}]] = \frac{2\sigma}{\sqrt{n}} [z_{1-\alpha}(1 - \alpha) + f(z_{1-\alpha})],$$

where Z is standard normal and f is the standard normal probability density function (PDF). In contrast, the length of the CI in eq. (1) is $2\sigma/\sqrt{n} \cdot z_{1-\alpha/2}$. Thus, when $\theta = \theta_1$, the efficiency of the CI in eq. (1) is

$$\frac{z_{1-\alpha}(1 - \alpha) + f(z_{1-\alpha})}{z_{1-\alpha/2}},$$

which equals 84.99% when $\alpha = 0.05$. Thus, the CI in eq. (1) is “nearly” efficient in the sense that it is not possible to improve upon it (in terms of expected length) by more than 15% even if we correctly “direct power” at the right θ .

Note that the interval in eq. (2) can be arbitrarily long if $\bar{X}_n$ is far from θ_1 . So this interval is not something that we’d want to use in practice: we get a CI that is slightly shorter when $\bar{X}_n$ is close to θ_1 at the expense of being very long when $\bar{X}_n$ is far from θ_1 . $\square$

3. INVERTING ASYMPTOTICALLY VALID TESTS

Instead of inverting tests that are valid in finite samples, we can also invert tests that have a large-sample justification. Here we consider inverting the trinity of likelihood-based tests: the Wald, Lagrange multiplier (LM) and likelihood ratio (LR) tests.

Let $X = (X_1, \dots, X_n)$ be a random sample from a distribution with density $f(x \mid \theta)$, $\theta \in \Theta$. Let $\mathcal{I}(\theta)$ denote the (per-observation) information matrix. Under regularity conditions,

$$\sqrt{n}(\hat{\theta}_{ML} - \theta) \Rightarrow \mathcal{N}(0, \mathcal{I}^{-1}(\theta)).$$

The delta method then implies that for any continuously differentiable function $g: \Theta \rightarrow \mathbb{R}$,

$$\sqrt{n}(g(\hat{\theta}_{ML}) - g(\theta)) \Rightarrow \mathcal{N}(0, \nabla g(\theta)' \mathcal{I}^{-1}(\theta) \nabla g(\theta)).$$

We can consistently estimate the asymptotic variance by, for example,

$$\hat{V} = \nabla g(\hat{\theta}_{ML})' \left(-\nabla^2 \ell_n(\hat{\theta}_{ML}) / n \right)^{-1} \nabla g(\hat{\theta}_{ML}).$$

Then, by Slutsky’s theorem,

$$\frac{\sqrt{n}(g(\hat{\theta}_{ML}) - g(\theta))}{\hat{V}^{1/2}} \Rightarrow \mathcal{N}(0, 1),$$

so that we can construct an approximate CI for $g(\theta)$ by inverting the Wald test, which yields the CI

$$g(\hat{\theta}_{ML}) \pm z_{1-\alpha/2} \hat{V}^{1/2} / \sqrt{n}. \quad (3)$$

The quantity $\hat{V}^{1/2} / \sqrt{n}$ is called the *standard error*.

Digression (Uniform asymptotic validity). The Wald test controls size in large samples, that is, for any null value $\theta \in \Theta$,

$$\lim_{n \rightarrow \infty} P_\theta(n(g(\hat{\theta}_{ML}) - g(\theta))^2 > \hat{V} z_{1-\alpha/2}^2) = \alpha.$$

Let $C(X) = [g(\hat{\theta}_{ML}) - z_{1-\alpha/2} \hat{V}^{1/2} / \sqrt{n}, g(\hat{\theta}_{ML}) + z_{1-\alpha/2} \hat{V}^{1/2} / \sqrt{n}]$ denote the Wald CI. The previous display implies that for all $\theta \in \Theta$, $\lim_{n \rightarrow \infty} P_\theta(g(\theta) \in C(X)) = 1 - \alpha$, and therefore the Wald CI achieves nominal coverage in large samples pointwise in θ ,

$$\inf_{\theta} \lim_{n \rightarrow \infty} P_\theta(g(\theta) \in C(X)) = 1 - \alpha.$$

However, it does not immediately follow that we can swap the limit and the infimum, so that

$$\lim_{n \rightarrow \infty} \inf_{\theta \in \Theta} P_\theta(g(\theta) \in C(X)) = 1 - \alpha, \quad (4)$$

i.e., that for n large enough, the level is approximately $1 - \alpha$. This stronger property means that the CI achieves nominal coverage in large samples *uniformly* in θ , not just pointwise. Without such a property, the asymptotically valid CI is not necessarily approximately valid for *any* n , no matter how large. While it is possible to show that eq. (4) holds under additional regularity conditions, there are important cases when it fails. $\boxtimes$

Example 3. Let $X = (X_1, \dots, X_n)$ be a random sample from a Bernoulli(θ) distribution. We want to construct a confidence set for the odds ratio $g(\theta) = \theta/(1 - \theta)$. We have $\hat{\theta}_{ML} = \bar{X}_n$, and

$$\sqrt{n}(\hat{\theta}_{ML} - \theta) \Rightarrow \mathcal{N}(0, \theta(1 - \theta)).$$

In addition,

$$g'(\theta) = \frac{(1 - \theta) + \theta}{(1 - \theta)^2} = \frac{1}{(1 - \theta)^2}.$$

By the delta method,

$$\sqrt{n}(g(\hat{\theta}_{ML}) - g(\theta)) \Rightarrow \mathcal{N}(0, \theta/(1 - \theta)^3).$$

We can use the plug-in estimator $\hat{V} = \hat{\theta}_{ML}/(1 - \hat{\theta}_{ML})^3$ of the asymptotic variance. This leads to an approximate CI for $\theta/(1 - \theta)$ given by

$$\left[\frac{\hat{\theta}_{ML}}{1 - \hat{\theta}_{ML}} - z_{1-\alpha/2} \sqrt{\frac{\hat{\theta}_{ML}}{(1 - \hat{\theta}_{ML})^3 n}}, \frac{\hat{\theta}_{ML}}{1 - \hat{\theta}_{ML}} + z_{1-\alpha/2} \sqrt{\frac{\hat{\theta}_{ML}}{(1 - \hat{\theta}_{ML})^3 n}} \right]. \quad (5) \quad \boxtimes$$

Digression (Uniform size control). Following up on the earlier digression, if we define the parameter space in the previous example as $\Theta = (0, 1)$, then none of the asymptotic methods actually achieves uniform asymptotic validity: for any n , no matter how large, we can find a θ close enough to 0 or 1 such that the CI will undercover. In practice, if one uses one of these asymptotic methods, one has to implicitly rule out these problematic coverage regions, that is, assume that $\Theta = [\epsilon, 1 - \epsilon]$ for some $\epsilon > 0$. For this parameter space, coverage is uniform. $\boxtimes$

3.1. Confidence Sets Based on LM and LR Tests

In addition to the Wald statistic, we can invert tests based on the LM and the LR statistics as well. However, these confidence sets are usually more involved. Under regularity conditions, the trinity of confidence sets is asymptotically equivalent.

Example 3 (continued). Let's first consider a CI for θ . The CI based on inverting a Wald test is given by

$$\hat{\theta}_{ML} \pm z_{1-\alpha/2} \sqrt{\hat{\theta}_{ML}(1 - \hat{\theta}_{ML})/n}.$$

To derive a CI based on inverting the LR test, note that the joint log-likelihood is

$$\ell_n(\theta) = \log \left(\theta^{\sum X_i} (1 - \theta)^{n - \sum X_i} \right) = n\bar{X}_n \log(\theta) + n(1 - \bar{X}_n) \log(1 - \theta).$$

Since $\hat{\theta}_{ML} = \bar{X}_n$,

$$\zeta_{LR}(\theta) = 2n [\bar{X}_n \log(\bar{X}_n/\theta) + (1 - \bar{X}_n) \log((1 - \bar{X}_n)/(1 - \theta))]$$

This yields the CI

$$\{\theta \in [0, 1]: 2n [\bar{X}_n \log(\bar{X}_n/\theta) + (1 - \bar{X}_n) \log((1 - \bar{X}_n)/(1 - \theta))]\leq z_{1-\alpha/2}^2\} \quad (6)$$

It is the solution to a nonlinear inequality.

To derive a CI based on inverting the LM test, note

$$\mathcal{S}_n(\theta) = n \left(\frac{\bar{X}_n}{\theta} - \frac{1 - \bar{X}_n}{1 - \theta} \right) = n \frac{\bar{X}_n - \theta}{\theta(1 - \theta)}, \quad \mathcal{I}(\theta) = \frac{1}{\theta(1 - \theta)}.$$

Thus,

$$\zeta_{LM}(\theta) = \mathcal{S}_n(\theta) \mathcal{I}(\theta)^{-1} \mathcal{S}_n(\theta) / n = \frac{n(\bar{X}_n - \theta)^2}{\theta(1 - \theta)}.$$

We know that $\zeta_{LM}(\theta) \Rightarrow \chi_1^2$. So, the confidence set based on inverting the LM test is

$$\begin{aligned} & \left\{ \theta \in [0, 1]: n(\bar{X}_n - \theta)^2 \leq z_{1-\alpha/2}^2 \theta(1 - \theta) \right\} \\ &= \left\{ \theta \in [0, 1]: (1 + z_{1-\alpha/2}^2/n)\theta^2 - (2\bar{X}_n + z_{1-\alpha/2}^2/n)\theta + \bar{X}_n^2 \leq 0 \right\}. \end{aligned} \quad (7)$$

It is the solution to a quadratic inequality.

What about CIs for $g(\theta) = \theta/(1 - \theta)$? The LR statistic is invariant to reparametrization. Further, since the restricted value of θ under the null $H_0: g(\theta) = g_0$ is $\theta_0 = g^{-1}(g_0) = g_0/(1 + g_0)$, the CI is given by

$$\{g_0 \in (0, \infty): \zeta_{LR}(g^{-1}(g_0)) \leq z_{1-\alpha/2}^2\}.$$

Similarly, for the LM test, since the restricted value of θ under the null is $g_0/(1 + g_0)$, the CI is given by

$$\begin{aligned} & \{g_0 \in (0, \infty): \zeta_{LM}(g^{-1}(g_0)) \leq z_{1-\alpha/2}^2\} \\ &= \left\{ g_0 \in (0, \infty): n(\bar{X}_n(1 + g_0) - g_0)^2 \leq z_{1-\alpha/2}^2 g_0 \right\}. \end{aligned}$$

Notice that for both the LM and LR sets, the confidence set is a *projection* of the confidence set for θ . That is, we could equivalently first construct a CI for θ based on eq. (6) or eq. (7), and then collect all values of $g(\theta)$ that fall inside this CI. We would end up with the

same result. However, projecting the Wald CI for θ yields the CI

$$\left[\frac{\hat{\theta}_{ML} - z_{1-\alpha/2} \sqrt{\hat{\theta}_{ML}(1 - \hat{\theta}_{ML})/n}}{1 - \hat{\theta}_{ML} + z_{1-\alpha/2} \sqrt{\hat{\theta}_{ML}(1 - \hat{\theta}_{ML})/n}}, \frac{\hat{\theta}_{ML} + z_{1-\alpha/2} \sqrt{\hat{\theta}_{ML}(1 - \hat{\theta}_{ML})/n}}{1 - \hat{\theta}_{ML} - z_{1-\alpha/2} \sqrt{\hat{\theta}_{ML}(1 - \hat{\theta}_{ML})/n}} \right],$$

which is different from the Wald CI for $g(\theta)$ we derived in eq. (5), because the Wald statistic is not invariant to reparametrization: a Wald test of the null $H_0: \theta = \theta_0$ may reject, but a Wald test of the null $H_0: g(\theta) = g(\theta_0)$ may not.

Note that when all trials are successes, all methods except Wald projection report intervals that are unbounded from above (that is, the upper endpoint is ∞). Wald projection reports $\{\infty\}$, because the standard error for θ is then zero.

Figure 1 plots the coverage rates of the four intervals (LM, LR, Wald, and projecting the Wald interval for θ) when $n = 200$. You can see that the coverage is quite jagged for all methods: the discreteness of the data still plays a role even at this sample size. Furthermore, coverage dips below the nominal level for all methods for extreme values of θ , though the problem is less severe for LM and LR. Figure 1 also plots the length of these intervals relative to the interval obtained by inverting the UMP unbiased test (which, since X is discrete, is randomized). Since the expected length for the odds ratio is infinite, we plot the expectation of the length trimmed at 500. We see that projecting the Wald test for θ to obtain a CI for the odds ratio does poorly in terms of expected length. While LM has the best coverage, it pays for it in terms of expected length. $\square$

4. BERNSTEIN-VON MISES

While the form of the confidence sets differs in small samples, one can show that CIs based on inverting Wald, LM and LR tests are all asymptotically equivalent. How do they compare to Bayesian credible intervals?

Theorem 2. Suppose that the conditions for asymptotic normality of $\hat{\theta}_{ML}$ hold. Consider the posterior distribution for the rescaled and recentered parameter $\vartheta = \sqrt{n}(\theta - \theta_0)$ under a continuous prior distribution $\Pi(\theta)$, such that the prior density $\pi(\theta)$ is continuous and positive at θ_0 . Then the posterior distribution for ϑ converges in probability to the distribution $\mathcal{N}(\sqrt{n}(\hat{\theta}_{ML} - \theta_0), \mathcal{I}(\theta_0)^{-1})$.

The convergence in the theorem is in total variation distance, that is $\sup_A |\Pi(\vartheta \in A \mid X) - P_{\mathcal{N}(\sqrt{n}(\hat{\theta}_{ML} - \theta_0), \mathcal{I}(\theta_0)^{-1})}(A)| \xrightarrow{P} 0$. This notion of convergence is needed because both the posterior distribution and the normal approximating distributions are random: the former depends on the data, and the latter depends on $\hat{\theta}_{ML}$, which is random. To talk about two random distributions being close in probability, we need to convert the difference between them into a single number, which is what the total variation distance does. If the two distributions are continuous with PDFs g_n and f_n , respectively, then the statement is equivalent to $\int |g_n(x) - f_n(x)| dx \xrightarrow{P} 0$.

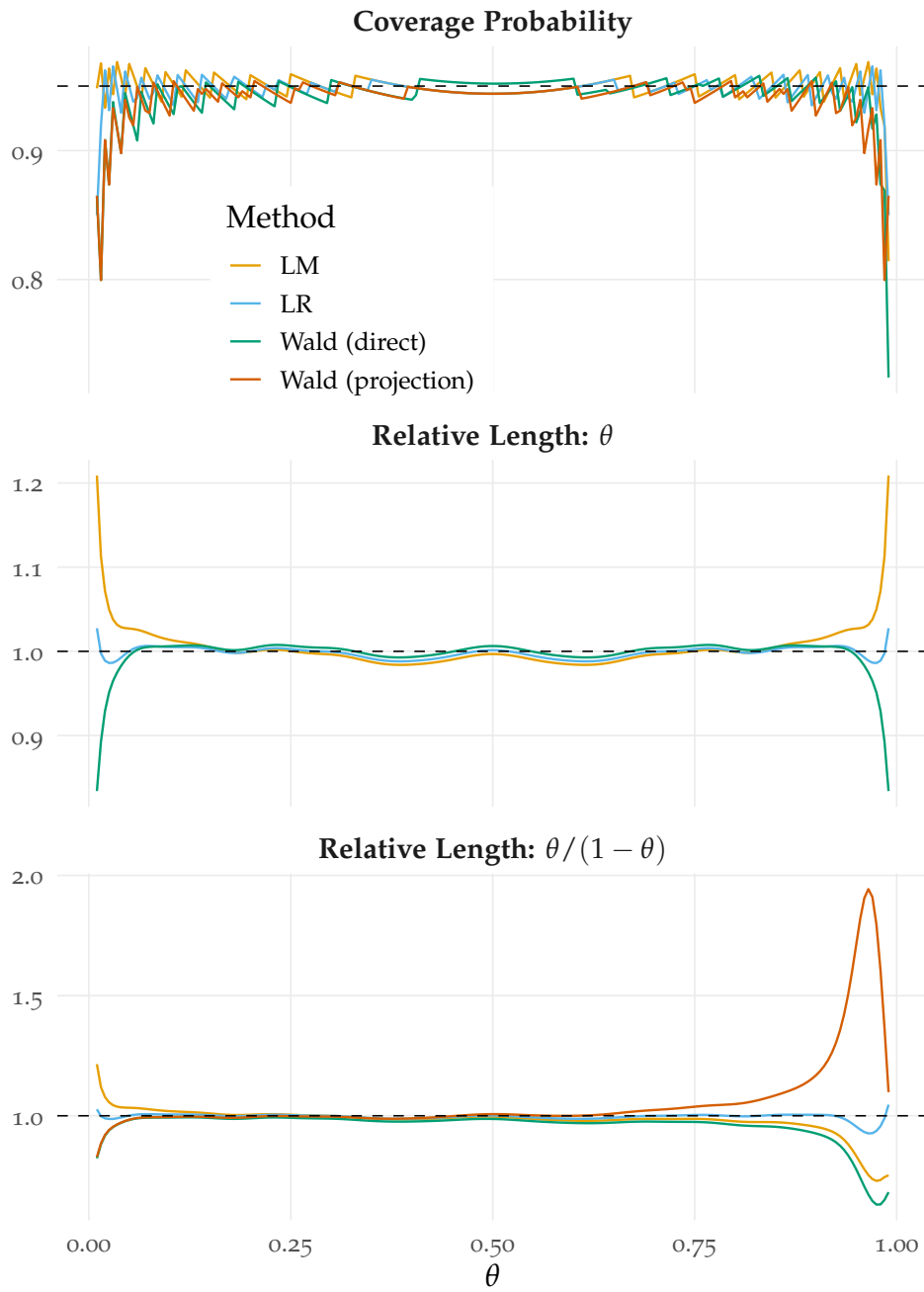


Figure 1: Coverage and expected length of LM, LR, Wald, and the projected Wald CIs relative to the CI based on inverting the UMP unbiased test, for $n = 200$. For the odds ratio parameter, expected length is trimmed at 500.

Proof. We prove the result pointwise, and show that the ratio of PDFs at a single point ϑ converges to one in probability. For the stronger result that the integral of the absolute values of the differences converges to zero in probability, see van der Vaart (1998, Theorem 10.1). By the change of variables formula, the posterior density for ϑ is given by $\pi(\theta_0 + \vartheta/\sqrt{n} \mid X)/\sqrt{n}$, where $\pi(\theta \mid X)$ is the posterior for θ . If we scale the posterior by the posterior at θ_0 , we get

$$\frac{\pi(\theta_0 + \vartheta/\sqrt{n} \mid X)}{\pi(\theta_0 \mid X)} = \frac{\mathcal{L}_n(\theta_0 + \vartheta/\sqrt{n})\pi(\theta_0 + \vartheta/\sqrt{n})}{\mathcal{L}_n(\theta_0)\pi(\theta_0)}.$$

Since π is continuous at θ_0 , the ratio of priors satisfies $\pi(\theta_0 + \vartheta/\sqrt{n})/\pi(\theta_0) \rightarrow 1$. Furthermore, the log-likelihood difference satisfies, by a Taylor expansion,

$$\ell_n(\theta_0 + \vartheta/\sqrt{n}) - \ell_n(\theta_0) = \vartheta' \mathcal{S}_n(\theta_0)/\sqrt{n} + \frac{1}{2n} \vartheta' \nabla^2 \ell_n \vartheta,$$

where each element of the second derivative $\nabla^2 \ell_n$ is evaluated at some intermediate value of θ that lies between θ_0 and $\theta_0 + \vartheta/\sqrt{n}$ (but the mean-value point may be different for different elements of $\nabla^2(\ell_n)$). By the usual hand-waving, since for a fixed θ , $\frac{1}{n} \nabla^2 \ell_n(\theta) \xrightarrow{p} -\mathcal{I}(\theta)$, this suggests

$$\ell_n(\theta_0 + \vartheta/\sqrt{n}) - \ell_n(\theta_0) - \vartheta' \mathcal{S}_n(\theta_0)/\sqrt{n} + \frac{1}{2} \vartheta' \mathcal{I}(\theta_0) \vartheta \xrightarrow{p} 0. \quad (8)$$

Let $\Delta_n = \sqrt{n}(\hat{\theta}_{ML} - \theta_0)$. If we evaluate the normal density $\mathcal{N}(\Delta_n, \mathcal{I}(\theta_0)^{-1})$ at a point ϑ , the log-density is,

$$\log p_{\mathcal{N}}(\vartheta) = C - \frac{1}{2} (\Delta_n - \vartheta)' \mathcal{I}(\theta_0) (\Delta_n - \vartheta),$$

where C is a constant. In the proof of asymptotic normality in Lecture 6, we saw that $\Delta_n - \mathcal{I}(\theta_0)^{-1} \mathcal{S}_n(\theta_0)/\sqrt{n} \xrightarrow{p} 0$. Hence,

$$\begin{aligned} \log p_{\mathcal{N}}(\vartheta) - \vartheta' \mathcal{S}_n(\theta_0)/\sqrt{n} + \frac{1}{2} \vartheta' \mathcal{I}(\theta_0) \vartheta &= C - \frac{1}{2} \Delta_n' \mathcal{I}(\theta_0) \Delta_n \\ &\quad + \vartheta' \mathcal{I}(\theta_0) [\Delta_n - \mathcal{I}(\theta_0)^{-1} \mathcal{S}_n(\theta_0)/\sqrt{n}], \end{aligned}$$

and the last term converges to zero in probability. Since the first two terms on the right-hand side of the previous display do not depend on ϑ , from the perspective of the posterior, they act as normalizing constants. It therefore follows from eq. (8) that the difference between the log-posterior and the log-density of $\mathcal{N}(\Delta_n, \mathcal{I}(\theta_0)^{-1})$ converges to zero in probability. $\square$

This is a remarkable result, for two reasons. First, the limit distribution does not depend on the prior: the prior gets dominated by the likelihood, since as $n \rightarrow \infty$, whatever one's prior beliefs are, they get dominated by the data. Loosely speaking, for n large, the posterior for θ is given by

$$\theta \mid X \approx \mathcal{N}(\hat{\theta}_{ML}, \mathcal{I}(\theta_0)^{-1}/n), \quad (9)$$

while in Lecture 6, we proved that $\hat{\theta}_{ML} \mid \theta \approx \mathcal{N}(\theta, \mathcal{I}(\theta_0)^{-1}/n)$.

The equivalence between the distributions arises because, by a Taylor expansion, in the neighborhood of the true θ_0 (for a fixed ϑ), the log-likelihood is approximately quadratic in large samples. In other words, it is approximately Gaussian. Because the prior for ϑ is approximately flat, the posterior is also approximately Gaussian. On the

other hand, as we saw in the proof of asymptotic normality of maximum likelihood estimator (MLE), $\sqrt{n}(\hat{\theta}_{ML} - \theta_0)$ will approximately equal the inverse information times the score, and the score is also Gaussian in large samples by the central limit theorem (CLT).

The second reason why Theorem 2 is remarkable is that as a consequence of eq. (9), whatever Bayes decision one makes based on the posterior, so long as the decision is sufficiently smooth in the posterior, the Bayes decision will in large samples converge to a Bayes rule based on eq. (9). In particular:

1. Bayes posterior mean and posterior median estimators for $g(\theta)$ will in large samples be equivalent to $g(\hat{\theta}_{ML})$. They inherit all the large-sample optimality properties of MLE.
2. Bayes credible sets will in large samples be equivalent to the Wald CI in eq. (3), and also equivalent to CIs based on inverting LM, or LR tests.

Therefore, from a frequentist perspective, Bayes procedures are optimal in large samples. Vice versa, a Bayesian reading frequentist papers can interpret likelihood-based CIs as approximate credible sets, and MLEs as approximate posterior means.

ONE-PARAGRAPH SUMMARY OF THIS COURSE In Lecture 7, we showed that in finite samples admissible rules are Bayes, and that the reverse is also true (with some qualifications regarding dogmatic priors and limits of Bayes rules). Theorem 2 tells us that Bayes rules are optimal in large samples, and equivalent to likelihood-based inference. In regular parametric models, there is no wedge in large samples between optimal frequentist and Bayesian methods. But Bayesian methods have the additional advantage that they are optimal in finite samples in terms of average risk. Furthermore, in regular parametric models, estimation and inference just boil down to inference on θ based on a single normal observation $\mathcal{N}(\theta, \mathcal{I}(\theta_0)^{-1}/n)$. This is the basis for the large-sample optimality statements about likelihood-based inference.

This raises the question of why bother with frequentist inference at all. In fully parametric models that are regular, there is arguably very little reason. But more generally, I would argue that there are two reasons for using frequentist methods, in addition to the reason that not all models are regular. The first is that often, making the model fully parametric is hard. For instance, in a regression, one may want to allow for clustered and heteroskedastic errors, but modeling the heteroskedasticity is hard business. As you'll see in the second half of the course, frequentists can appeal to large-sample results to avoid such modeling. The second reason is that we are often worried about model misspecification, and we want to make sure that our procedures are interpretable even if the model is wrong. In fact, this is in my view the main reason why linear regression is useful and should be used—more on that in 539B. In many models of interest, it is not that difficult to construct misspecification-robust CIs. In contrast, allowing for misspecification in a Bayesian world can be a bit more involved.

5. (OPTIONAL) EXCITING NEW DEVELOPMENTS

This course has covered “standard” results in statistical theory. There are many recent results that are quite exciting, and that think about statistics in a very different way. I will give three examples below. If you manage to figure out how to make novel use of these methods in econometrics, you’ll have a thesis topic sorted out!

1. E-values. For instance, <https://doi.org/10.1214/23-STS894>
2. The HulC <https://doi.org/10.1093/jrsssb/qkad134> and universal inference <https://doi.org/10.1073/pnas.1922664117>
3. Conformal prediction. For instance, <https://doi.org/10.1007/978-3-031-06649-8>

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